

Easy (10 questions)

1.

What does IP stand for?

- A) Internet Process
- B) Internet Protocol
- C) Internal Protocol
- D) Interface Protocol

Answer: B

Explanation: IP stands for Internet Protocol, used for addressing and routing packets.

2.

Which layer of the OSI model does IP operate on?

- A) Data Link
- B) Transport
- C) Network
- D) Application

Answer: C

Explanation: IP operates at the Network Layer (Layer 3) of the OSI model.

3.

Which IP version uses 32-bit addresses?

- A) IPv2
- B) IPv4
- C) IPv6
- D) IPv8

Answer: B

Explanation: IPv4 uses 32-bit addresses.

4.

Which protocol is connection-oriented?

- A) TCP
- B) UDP
- C) IP
- D) ICMP

Answer: A

Explanation: TCP establishes a connection and ensures reliable data delivery.

5.

What does TTL stand for in IP packets?

- A) Time to Listen
- B) Total Transmission Length
- C) Time to Live
- D) Transport to Link

Answer: C

Explanation: TTL limits the lifespan of a packet to prevent infinite routing loops.

6.

Which TCP/IP layer is responsible for routing?

- A) Application
- B) Internet
- C) Transport
- D) Network Access

Answer: B

Explanation: The Internet layer handles routing and logical addressing.

7.

Which protocol translates domain names to IP addresses?

- A) FTP

- B) DNS
- C) ARP
- D) ICMP

Answer: B

Explanation: DNS resolves human-readable domain names into IP addresses.

8.

UDP is a:

- A) Connection-oriented protocol
- B) Connectionless protocol
- C) Routing protocol
- D) Physical layer protocol

Answer: B

Explanation: UDP is connectionless and does not guarantee delivery.

9.

Which TCP/IP layer corresponds to OSI's Physical and Data Link layers?

- A) Internet
- B) Network Access
- C) Transport
- D) Application

Answer: B

Explanation: The Network Access layer manages physical transmission and network hardware.

10.

What does ARP do?

- A) Resolves IP addresses to MAC addresses
- B) Resolves MAC addresses to IP addresses
- C) Routes packets across networks
- D) Provides encryption

Answer: A

Explanation: ARP maps IP addresses to MAC addresses within a local network.

Medium (30 questions)

11.

Which field in the IPv4 header specifies the protocol of the encapsulated payload?

- A) TTL
- B) Protocol
- C) Source IP
- D) Flags

Answer: B

Explanation: The Protocol field indicates the next level protocol (e.g., TCP, UDP).

12.

What is the size of an IPv6 address?

- A) 32 bits
- B) 64 bits
- C) 128 bits
- D) 256 bits

Answer: C

Explanation: IPv6 addresses are 128 bits long.

13.

Which of the following is NOT a characteristic of TCP?

- A) Reliable delivery
- B) Ordered data transfer
- C) Connectionless
- D) Flow control

Answer: C

Explanation: TCP is connection-oriented, not connectionless.

14.

What is the default subnet mask for a Class C IPv4 address?

- A) 255.0.0.0
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.255.255.255

Answer: C

Explanation: Class C uses 255.255.255.0 as the default subnet mask.

15.

What is the primary purpose of the ICMP protocol?

- A) Data transfer
- B) Error reporting and diagnostics
- C) Address resolution
- D) Domain name resolution

Answer: B

Explanation: ICMP is used for error reporting and network diagnostics like ping.

16.

Which layer in TCP/IP model is responsible for segmentation and flow control?

- A) Internet
- B) Network Access
- C) Transport

D) Application

Answer: C

Explanation: The Transport layer handles segmentation, flow control, and error handling.

17.

How many bits are in an IPv4 address?

A) 8

B) 16

C) 32

D) 64

Answer: C

Explanation: IPv4 addresses consist of 32 bits.

18.

Which IP address is a loopback address?

A) 127.0.0.1

B) 192.168.1.1

C) 10.0.0.1

D) 255.255.255.255

Answer: A

Explanation: 127.0.0.1 is the loopback address used for testing.

19.

What is fragmentation in IP?

A) Splitting large packets into smaller ones

B) Encrypting packets

C) Routing packets to correct destination

D) Checking packet integrity

Answer: A

Explanation: Fragmentation splits large packets to fit smaller MTUs.

20.

Which of the following is true about UDP?

- A) Provides guaranteed delivery
- B) Uses acknowledgments
- C) Is suitable for streaming applications
- D) Uses a three-way handshake

Answer: C

Explanation: UDP is fast and suitable for streaming where speed is prioritized over reliability.

21.

Which field in the IPv4 header indicates fragmentation information?

- A) Identification
- B) Flags
- C) Fragment Offset
- D) All of the above

Answer: D

Explanation: These fields collectively manage fragmentation.

22.

What is the maximum TTL value in an IP packet?

- A) 64
- B) 128
- C) 255
- D) 512

Answer: C

Explanation: TTL is an 8-bit field, max value 255.

23.

What does CIDR stand for?

- A) Classful Inter-Domain Routing
- B) Classless Inter-Domain Routing
- C) Computer Inter-Domain Routing
- D) Common Internet Domain Routing

Answer: B

Explanation: CIDR allows flexible IP address allocation by removing class boundaries.

24.

Which protocol is used to find the MAC address associated with an IP address?

- A) DNS
- B) ICMP
- C) ARP
- D) TCP

Answer: C

Explanation: ARP resolves IP addresses to MAC addresses.

25.

In TCP/IP, which protocol provides reliable end-to-end delivery?

- A) IP
- B) UDP
- C) TCP
- D) ICMP

Answer: C

Explanation: TCP ensures reliable data transmission with acknowledgments and retransmissions.

26.

What is the size of the IPv4 header without options?

- A) 20 bytes
- B) 24 bytes
- C) 40 bytes

D) 60 bytes

Answer: A

Explanation: The IPv4 header is 20 bytes when no options are included.

27.

Which protocol does the “ping” utility use?

A) TCP

B) UDP

C) ICMP

D) ARP

Answer: C

Explanation: Ping uses ICMP echo request and reply messages.

28.

What is the primary purpose of the Network Access layer in TCP/IP?

A) Routing

B) Data encapsulation and physical transmission

C) Segmentation

D) Application services

Answer: B

Explanation: It deals with hardware addressing and physical data transfer.

29.

Which layer of TCP/IP handles encryption and decryption?

A) Application

B) Transport

C) Internet

D) Network Access

Answer: A

Explanation: Encryption is usually handled at the Application layer (e.g., HTTPS).

30.

Which of the following is NOT a protocol in the TCP/IP suite?

- A) FTP
- B) SMTP
- C) IPX
- D) DNS

Answer: C

Explanation: IPX is a protocol used in Novell networks, not TCP/IP.

Hard (10 questions)

31.

How does IPv6 handle fragmentation differently from IPv4?

- A) Routers perform fragmentation
- B) Only the sender performs fragmentation
- C) No fragmentation is allowed
- D) IPv6 does not support fragmentation

Answer: B

Explanation: In IPv6, only the source node performs fragmentation; routers do not fragment packets.

32.

What is the significance of the "Protocol" field in the IPv4 header?

- A) Indicates the protocol to be used by the IP layer
- B) Identifies the encapsulated transport layer protocol
- C) Specifies encryption method

D) Determines fragmentation rules

Answer: B

Explanation: It tells the IP layer what protocol is encapsulated in the payload (e.g., TCP, UDP).

33.

Which TCP flag is used to initiate a connection?

A) SYN

B) ACK

C) FIN

D) RST

Answer: A

Explanation: The SYN (synchronize) flag is used to initiate a TCP connection.

34.

In the TCP three-way handshake, which sequence correctly describes the process?

A) SYN → ACK → SYN-ACK

B) SYN → SYN-ACK → ACK

C) ACK → SYN → SYN-ACK

D) SYN-ACK → SYN → ACK

Answer: B

Explanation: The handshake is SYN (client), SYN-ACK (server), ACK (client).

35.

What is the purpose of the “Identification” field in the IPv4 header?

A) To identify the packet’s sender

B) To assist in reassembling fragmented packets

C) To identify the routing protocol

D) To specify QoS level

Answer: B

Explanation: The Identification field helps in reassembling fragments of a packet.

36.

Which of the following best describes the "Checksum" field in the IPv4 header?

- A) Verifies the entire packet including payload
- B) Verifies only the IP header integrity
- C) Verifies the payload integrity
- D) Encrypts the IP header

Answer: B

Explanation: The checksum verifies the IP header only.

37.

Which mechanism in TCP controls the rate of data transmission to prevent network congestion?

- A) Flow Control
- B) Congestion Control
- C) Error Detection
- D) Fragmentation

Answer: B

Explanation: TCP uses congestion control algorithms to avoid overloading the network.

38.

What is the effect of the "Don't Fragment (DF)" flag in an IPv4 packet?

- A) Packet must be fragmented
- B) Packet cannot be fragmented
- C) Packet is discarded if fragmentation needed
- D) Packet is duplicated

Answer: B

Explanation: The DF flag instructs routers not to fragment the packet.

39.

Which TCP segment flag indicates the termination of a connection?

- A) SYN
- B) ACK
- C) FIN
- D) PSH

Answer: C

Explanation: The FIN (finish) flag is used to terminate a TCP connection.

40.

What is the maximum number of hosts possible on a Class C network without subnetting?

- A) 254
- B) 256
- C) 512
- D) 1024

Answer: A

Explanation: Class C supports 256 addresses minus network and broadcast addresses = 254 hosts.

41.

In IPv6, what is the function of the "Next Header" field?

- A) Specifies the next packet to be sent
- B) Indicates the type of header immediately following the IPv6 header
- C) Used for fragmentation
- D) Indicates encryption algorithm

Answer: B

Explanation: The Next Header field indicates the type of the next header after the IPv6 header.

42.

How does the TCP Sliding Window mechanism work?

- A) Sends data one byte at a time
- B) Controls the number of bytes that can be sent before receiving an acknowledgment
- C) Limits IP fragmentation

D) Encrypts data during transmission

Answer: B

Explanation: The sliding window controls flow by allowing a certain amount of unacknowledged data.

43.

Which protocol is used to dynamically assign IP addresses to hosts?

- A) DNS
- B) DHCP
- C) ICMP
- D) FTP

Answer: B

Explanation: DHCP (Dynamic Host Configuration Protocol) assigns IP addresses dynamically.

44.

Which of the following IPv4 addresses is reserved for multicast?

- A) 192.168.0.0/16
- B) 224.0.0.0/4
- C) 127.0.0.0/8
- D) 169.254.0.0/16

Answer: B

Explanation: 224.0.0.0 to 239.255.255.255 is the multicast address range.

45.

What is the purpose of the SYN flood attack?

- A) To disrupt IP routing
- B) To exhaust server resources by sending many connection requests without completing handshake
- C) To steal IP addresses
- D) To encrypt packets

Answer: B

Explanation: SYN flood attacks overwhelm servers by initiating many TCP connections without completing them.

46.

What is the use of the "Urgent Pointer" field in TCP header?

- A) Indicates urgent data within the segment
- B) Specifies connection priority
- C) Identifies source port
- D) Specifies encryption key

Answer: A

Explanation: It points to urgent data that should be processed immediately.

47.

How does IPsec enhance the IP protocol?

- A) Adds encryption and authentication at the IP layer
- B) Improves routing efficiency
- C) Allows fragmentation
- D) Handles domain name resolution

Answer: A

Explanation: IPsec provides security features like encryption and authentication.

48.

Which of the following best describes a broadcast IP address?

- A) Address used to send data to all hosts in a subnet
- B) Address used to send data to a single host
- C) Address used to route packets between networks
- D) Address used for IPv6 only

Answer: A

Explanation: Broadcast addresses send data to all hosts in a network segment.

49.

Which TCP flag can be used to reset a connection?

- A) SYN
- B) RST
- C) FIN
- D) PSH

Answer: B

Explanation: The RST (reset) flag is used to abruptly terminate a connection.

50.

Which of the following is NOT a feature of the IPv6 header?

- A) Simplified header structure
- B) No checksum field
- C) Variable header length
- D) Larger address space

Answer: C

Explanation: IPv6 has a fixed header length (40 bytes), unlike IPv4.